A. K. M. SHAHADAT HOSSAIN

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PROFESSIONAL SUMMARY

Computer Science & Engineering (CSE) student at BRAC University with experience developing machine learning, computer vision, and full-stack web applications. Built projects involving object detection, deep learning ensembles, explainable AI, and Full-stack development. Seeking opportunities to apply my skills in real-world environments and build a career in AI/ML Engineering and Software Engineering.

EDUCATION

B.Sc. in Computer Science & Engineering (CSE)
BRAC University | CGPA: 3.59 /4.00 2022 - 2026

Higher Secondary Certificate (HSC)
St. Joseph Higher Secondary School | GPA: 5.00 / 5.00 2019 - 2021

Secondary School Certificate (SSC)
St. Joseph Higher Secondary School | GPA: 5.00 / 5.00 2011 - 2019

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, x86 Assembly, HTML, CSS

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, convolutional neural networks, transfer learning, model ensembling, object detection (YOLOv8), image classification, explainable AI (Grad-CAM), out-of-distribution detection, model training, evaluation and inference optimization

Web Development: React.js, Node.js, Express.js, TypeScript, REST API, JWT authentication, MongoDB, MySQL, Tailwind CSS, Supabase PostgreSQL, Supabase Auth

Tools & Practices: Git/GitHub, Linux, Bash, Virtual environments, Jupyter, PowerBI, Agile/Scrum, Version control workflows, Supabase

Engineering & Research Methods: Data structures & algorithms, Object-oriented programing, Operating systems, Computer networks, Databases management systems, Software engineering, System design & development, Experiment design, Agile methodology

SELECTED PROJECTS

Ensemble CNN for Sugar Adulteration Detection with OOD Rejection & Grad-CAM
Undergraduate thesis, BRAC University - Supervisor: Dr. Chowdhury Mofizur Rahman *[May 2025 – April 2026]*
Stack: Python, PyTorch, ResNet50, DenseNet121, ConvNeXt-Tiny, scikit-learn, NumPy, Pandas

- Designed a three-backbone ensemble (ResNet50, DenseNet121, ConvNeXt-Tiny) that classifies adulterated vs. pure sugar from ordinary smartphone photos at 96.3% accuracy, versus 90.6% for the strongest single backbone.
- Built a Mahalanobis-distance-based out-of-distribution detector over penultimate-layer features to identify unfamiliar sugar adulterants, achieving a mean Mahalanobis distance of 90.69 for correctly classified samples and 88.07 for incorrectly classified samples, with OOD evaluation performed on previously unseen urea and saccharine adulterants.
- Integrated Grad-CAM attribution to show which image regions drove each prediction, converting a black-box classifier into an output a non-ML reviewer can audit.
- Owned the full pipeline: dataset capture and labelling (3200 images across 5 classes), augmentation, transfer learning and fine-tuning, stratified 5-fold evaluation, and reproducible training scripts.

- Replaces reliance on laboratory chemical assays and dedicated spectrometry hardware with smartphone RGB imaging, making non-destructive adulteration screening more affordable and accessible to consumers.
- Source: https://github.com/shehedet/Ensemble-CNN-for-Sugar-Adulteration-Detection-with-OOD-Rejection-Grad-CAM

SeeMenu: A Community-driven Restaurant and Food Menu Discovery Platform *[Oct 2026 - Ongoing]*
Stack: React.js, Vite, Tailwind CSS, Node.js, Express.js, REST APIs, Supabase PostgreSQL, Supabase Auth

- Developed a work-in-progress full-stack restaurant discovery platform, with a responsive React interface, search and filtering, and a Node.js/Express backend.
- Integrated Supabase authentication with email verification, login/logout flows, and role-aware routing for Contributor, Moderator, and Admin roles.
- Implemented landing-page search behavior, empty-result handling, and the Express API health-check endpoint.
- Designed an initial PostgreSQL schema spanning nine tables for user profiles, restaurants, menu images, comments, voting, reports, restaurant requests, and notifications.
- Built the front end as reusable React components with client-side routing and Tailwind styling.
- Implemented database-backed restaurant search, menu image uploads, secure role-based permissions, moderation workflow, automated testing, live deployment
- Source: https://github.com/shehedet/SeeMenu
- Usable link: https://see-menu.vercel.app

Veggie-Valley: Full-Stack Marketplace for Local Produce *[Aug 2025 – Sep 2025]*
Stack: React.js, Node.js, Express.js, MongoDB, REST APIs, JWT, Supabase, Tailwind CSS

- Shipped a full-stack marketplace where growers list produce and buyers browse, order and pay, owning the complete path from schema design to deployed UI.
- Designed and implemented REST APIs for authentication, catalog, cart and order lifecycle, with JWT-based sessions and role-based authorization separating buyer and seller permissions.
- Modelled the product, user and order collections in MongoDB and used Supabase for image storage.
- Integrated a sandbox payment gateway and built the checkout state machine (cart → payment → order confirmation → order history), including failure and retry paths.
- Built the front end as reusable React components with client-side routing and Tailwind styling.
- Source: https://github.com/shehedet/Veggie-Valley
- Usable link: https://veggie-valley-final.vercel.app

Aerial Vehicle & Pedestrian Detection and Counting Pipeline *[May 2026]*
Stack: Python, PyTorch, YOLOv8s, OpenCV, NumPy, Linux, Git

- Built an end-to-end computer vision pipeline that ingests drone footage, runs YOLOv8s object detection, tracks objects across frames, and returns per-class counts of vehicles and pedestrians.
- Trained and evaluated the detector on 8629 annotated aerial images, reaching 92% mAP@0.5 and sustaining 75 FPS real-time inference on NVIDIA RTX 4050.
- Handled the hardest failure mode in aerial data - small, densely packed objects at varying altitudes - through targeted augmentation and anchor/resolution tuning, raising recall on small objects from [X]% to [Y]%.
- Implemented frame-to-frame identity tracking so each object is counted once rather than once per frame.
- Source: https://github.com/shehedet/Vehicle-and-Human-Detection-and-Counting-System-using-Drone-Images

CERTIFICATIONS

- **Machine Learning Specialization** - DeepLearning.AI, Stanford Online, Coursera (Supervised Learning; Advanced Learning Algorithms; Unsupervised Learning, Recommenders & Reinforcement Learning).
- **Full-Stack Web Development** - IBM, Coursera (Developing Front-End Apps with React.js; Developing Back-End Apps with Node.js and Express.js).

EXTRACURRICULAR ACTIVITIES

- **Duke of Edinburgh's International Award, Bronze - BRAC University:** completed a multi-month program of skill development, service and an expedition, assessed on consistency and follow-through.